1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist synthesis, AR-targeted or chemo-primed tumors often show biomarker shifts (Decipher GC, PSA kinetics, IO predictive signatures) yet some patients fail to convert those signals into durable survival gains — i.e., why do biomarker-rich, therapy-primed tumors sometimes not achieve improved overall survival? 2) Tier 1 paths used: I built on Core Path 1 (Decipher GC linking therapy to long-term outcomes) and Core Path 3 (IO predictive signatures as a therapy-modulated immune readout). These directly connect tumor-intrinsic genomic risk (Decipher) and tumor-immune readouts (IO signatures) to clinical outcomes after AR-directed or neoadjuvant therapy. 3) Tier 2 support: CENPA, Sox2/NE programs, and inflammation (EDIH) provide mechanistic routes by which tumor-intrinsic aggressiveness (chromosomal instability, lineage plasticity) or pro/anti-inflammatory states could disable effective immune organization; PSA trajectories offer an orthogonal, therapy-responsive kinetic readout that can help time when immune dysfunction emerges. 4) Why OS: Overall survival is the most definitive, clinically meaningful endpoint that integrates metastatic progression and treatment resistance — appropriate for testing whether biomarker discordance predicts failure to translate biomarker signals into durable benefit. 5) Deliberate exclusions: I excluded exploratory/indirect paths (SPC, alpha-fetoprotein, neurocognitive uptake) from the core claim because they are distal to the tumor–immune–genomic axis and would dilute mechanistic focus. I limited the hypothesis to one imaging feature and two anchor molecular features to meet the Focusing Rule and keep the question testable and mechanistic?

Working Hypothesis

In localized high-risk prostate cancer, the presence of TLS-like lymphoid aggregates indicates a potential anti-tumor immune microenvironment, but high intrinsic tumor aggressiveness captured by Decipher GC reverses or blunts TLS-driven anti-tumor activity by promoting immune suppression/exhaustion, leading IO predictive signatures to be non-predictive in TLS-positive, Decipher-high tumors. Consequently, TLS-positive tumors with high Decipher risk and discordant IO signatures will not exhibit improved OS versus TLS-negative tumors, indicating that high Decipher risk overrides beneficial TLS/IO signals and identifying patients unlikely to derive durable benefit from IO-modulated therapies.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	Decipher GC, IO predictive signatures
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

Decipher GC: Decipher GC, PSA kinetics, IO predictive signatures) yet some patients fail to convert those signals into durable survival gains — i.

IO predictive signatures: IO predictive signatures) yet some patients fail to convert those signals into durable survival gains — i.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	PSA trajectories	Supporting modifier from knowledge graph
Secondary	CENPA	Supporting modifier from knowledge graph
Secondary	Sox2 / NE features	Supporting modifier from knowledge graph
Secondary	Inflammation biomarkers (EDIH)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PSA trajectories: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CENPA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.